

Software Development Organization

2026-2027

1 Project 1

1.1 Context

In a previous project, the Library Management REST-oriented backend service was developed.

This application provides REST endpoints for managing:

- Books
- Genres
- Authors
- Readers
- Lendings

1.2 Problem

The Library Management service has been developed and provides the required functionality, but its software development and delivery process has limitations regarding:

- automation and repeatability of build, testing, integration, and deployment activities;
- systematic assessment of software and test quality;
- early detection of defects and regressions;
- configuration and deployment across different environments;
- continuous integration and continuous delivery/deployment practices.

The development process should therefore evolve towards an automated, repeatable, and measurable CI/CD process that provides timely feedback about the quality and deployability of the software.

1.3 Goals

The goals of this project (P1) are:

- Assess the current development process and system, documenting the relevant *system-as-is*, including:
 - implementation and deployment;
 - existing build and testing practices;
 - test quantity, coverage, and effectiveness;
 - relevant software quality indicators.

- Design the target development and deployment process, documenting the relevant *system-to-be* and the decisions supporting its design.
- Implement an automated CI/CD pipeline that supports:
 - building and packaging the software;
 - static code analysis;
 - automated testing at appropriate levels;
 - test coverage and mutation testing;
 - reporting of build, quality, and test results;
 - creation and management of deployable artifacts;
 - deployment to different environments.
- Improve the automated testing strategy, considering different scopes and levels of testing and evaluating the ability of the tests to detect defects and regressions.
- Support configurable and repeatable deployment of the application across development, staging, and production environments.
- Evaluate and improve the CI/CD process, using collected evidence and appropriate metrics to:
 - compare the initial and improved development processes;
 - identify limitations and bottlenecks;
 - justify improvements introduced throughout the project.

1.4 Conditions

- Number of students per group:
 - 2
 - exceptionally 3 or 1, if accepted by faculty, and implies differences in goals
- First evaluation:
 - Mandatory
 - Submission deadline: 2026-11-01
 - Mode: Individual, synchronous, face-to-face
 - Weight: 100% (or 30% if second evaluation takes places)
- Second evaluation:
 - Optional
 - Submission deadline: (as P2 submission deadline)
 - Mode: Individual, synchronous, face-to-face
 - Weight: 30% (or 0% if it does not take place)

2 Project 2

TBD